EDC · MOCK

EDC mock A

Answers and worked solutions · Day 27 · Thu 19 Nov · pack EX1

Marking guide

Before you open any solution: write your raw mark per question in the table at the end. Award a mark only if your script shows the line that earns it. If your formula and substitution are right but an arithmetic slip gives a wrong number, take the method marks and lose the answer mark. Accept answers within about 2% of the model answer when the difference comes from rounding.

Where the questions come from. Three parts are workshop-style variants (same method, changed numbers), because the course reuses workshop questions: **Q2 is WS3 Q3**, **Q8(a) is WS10 Q2** and **Q10(b) is WS10 Q10**. Everything else is new. Every number below was computed in Python.

Suggested timing (82 marks, 120 min):

Q	Topic	Marks	Time	Clock
1	MCQ	10	8 min	0:08
2	Diode string design	8	11 min	0:19
3	BJT VDB, DC and AC	13	18 min	0:37
4	Emitter follower	9	13 min	0:50
5	JFET self-bias	8	12 min	1:02
6	E-MOSFET bias	7	10 min	1:12
7	Two-stage cascade	8	12 min	1:24
8	CMRR and diff amp	7	11 min	1:35
9	Op-amps	5	6 min	1:41
10	Power amps	7	10 min	1:51
	Check		9 min	2:00

Question 1: MCQ [10 marks]

Part	Answer	Reason
I	**(a)**	More reverse bias uncovers more fixed ion charge, so W grows. The junction behaves like a capacitor with a wider gap, so C_T falls (this is how a varactor works).
II	**(b)**	$V_r = V_P/(fCR)$, so ripple is inversely proportional to C. Doubling C roughly halves it. It never reaches zero.
III	**(c)**	The smallest $I_S = (V_{i,min} - V_Z)/R_S$, with most of it taken by the largest I_L, leaves the least for I_Z. If I_Z falls below I_{ZK}, regulation is lost. Case (a) is the worst case for Zener *power*, not for dropout.
IV	**(c)**	$V_{BC} = +0.5$ V means the collector-base junction is forward biased as well. Both junctions are forward biased, so the BJT is saturated. Check: $V_{CE} = V_{BE} - V_{BC} = 0.2$ V $= V_{CE,sat}$.
V	**(b)**	A capacitor is open at DC, so the bias is unchanged. At signal frequency it shorts R_E, so $\lvert A_v \rvert$ goes from $\approx R_C/(r_e + R_E)$ to $\approx R_C/r_e$. Z_i *falls* (from

Part	Answer	Reason
		$\beta(r_e + R_E)$ to βr_e), so (c) is wrong.
VI	**(a)**	At $V_{GS} = V_P$, Shockley gives $I_D = I_{DSS}(1-1)^2 = 0$. The channel is pinched off.
VII	**(c)**	A D-MOSFET has a built-in channel. It works in depletion mode ($V_{GS} < 0$) and in enhancement mode ($V_{GS} > 0$). A JFET must keep $V_{GS} \leq 0$, and an E-MOSFET needs $V_{GS} > V_{GS(Th)}$.
VIII	**(b)**	$g_m = g_{m0}(1 - V_{GS}/V_P)$ is a straight line falling from g_{m0} at $V_{GS} = 0$ to zero at V_P. At $I_D = I_{DSS}/4$, $g_m = 0.5 g_{m0}$.
IX	**(a)**	Virtual ground: high open-loop gain forces $V_- \approx V_+ = 0$, and the ideal input draws no current.
X	**(d)**	Each transistor needs about 0.7 V of V_{BE} before it conducts. Near zero crossing neither one conducts. The fix is to bias both slightly on with diodes (class AB).

MARKS: 1 per correct letter, no part marks.

COMMON MISTAKES: III: picking (a) because it is the worst case for P_Z. V: thinking the bypass raises Z_i. VII: choosing the JFET.

REREAD: E1 (I), E3 (II, III), E4–E5 (IV), E7 (V), E9 (VI, VII), E11 (VIII), E13 (IX, X).

Question 2: Diode string design [8 marks] (WS3 Q3 variant)

(a) Four equal diodes share 2.6 V, so each one drops

$$V_D = \frac{2.6}{4} = 0.65 \text{ V}$$

That is 0.05 V below 0.7 V, which is half a decade below 1 mA:

$$I_D = 1 \text{ mA} \times 10^{(0.65-0.7)/0.1} = 10^{-0.5} \text{ mA} = \mathbf{0.316 \text{ mA}}$$

$$R = \frac{V_{DD} - V_o}{I_D} = \frac{5 - 2.6}{0.316 \text{ mA}} = \mathbf{7.59 \text{ k}\Omega}$$

(b) Use the same diode law, $V_o = 4\,[0.7 + 0.1 \log_{10}(I_D/1 \text{ mA})]$, and KCL at the output: $I_D = (5 - V_o)/R - V_o/R_L$.

The load takes about 0.25 mA of the 0.32 mA, so the diode current falls to roughly a third. That is about half a decade, so each diode drops about 0.05 V and V_o falls by about 0.2 V. Start the iteration at $V_o = 2.4$ V:

Iteration	$I_R = (5 - V_o)/R$	$I_L = V_o/R_L$	$I_D = I_R - I_L$	New V_o
1 (V_o = 2.400 V)	0.3426 mA	0.2400 mA	0.1026 mA	2.4044 V
2 (V_o = 2.4044 V)	0.3420 mA	0.2404 mA	0.1016 mA	2.4027 V

$$\mathbf{V_o \approx 2.40 \text{ V}}, \qquad \mathbf{I_D \approx 0.102 \text{ mA}}$$

(The converged solution is $V_o = 2.403$ V and $I_D = 0.1018$ mA.) A starting guess of 2.6 V also converges, but it oscillates for several steps first.

(c) "0.1 V per decade" means $2.3\,nV_T = 0.1$ V, so $nV_T = 0.1/\ln 10 = 43.4$ mV.

$$r_d = \frac{nV_T}{I_D} = \frac{43.4 \text{ mV}}{0.316 \text{ mA}} = \mathbf{137\ \Omega} \text{ per diode}, \qquad 4r_d = 549\ \Omega$$

Small-signal divider:

$$\Delta V_o = \Delta V_{DD} \frac{4r_d}{R + 4r_d} = \pm 0.5 \times \frac{549}{7589 + 549} = \mathbf{\pm 33.7 \text{ mV}}$$

(Solving the exponential model exactly at 5.5 V gives $V_o = 2.631$ V, a change of 31 mV. The estimate is good.)

Marks (8):

- a. 1: 0.65 V per diode. 1: current of 0.316 mA from the decade rule. 1: $R = 7.59$ kΩ.
- b. 1: KCL with the load current written correctly. 1: a valid iteration (at least two steps). 1: $V_o \approx 2.40$ V.
- c. 1: $r_d = 137\ \Omega$, with nV_T found from the 0.1 V/decade figure. 1: ± 34 mV.

COMMON MISTAKES:

- Using 0.7 V per diode and then trying to "fit" 2.6 V. The decade rule is the whole point of the question.
- Using the small-signal estimate in (b). The load current is not small: the diode current changes by a factor of 3, so linearising gives 2.47 V instead of 2.40 V.
- Using $r_d = 26\ \mathrm{mV}/I_D$. Here $n \neq 1$, and the 0.1 V/decade data gives $nV_T = 43.4$ mV.

REREAD: E2 (multi-diode circuits, the decade rule), E1 (r_d).

Question 3: BJT voltage-divider bias, DC then AC [13 marks]

(a) Approximate method check: $\beta R_E = 120 \times 1\ \text{k}\Omega = 120\ \text{k}\Omega \geq 10R_2 = 82\ \text{k}\Omega$. The approximate method *may* be used, but the question asks for the exact method.

$$V_{Th} = \frac{R_2}{R_1 + R_2} V_{CC} = \frac{8.2}{47.2} \times 18 = \mathbf{3.127\ V}, \qquad R_{Th} = 39\ \text{k} \parallel 8.2\ \text{k} = \mathbf{6.775\ k\Omega}$$

$$I_B = \frac{V_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{3.127 - 0.7}{6.775\ \text{k} + 121\ \text{k}} = \mathbf{19.0\ \mu A}$$

$$I_C = \beta I_B = \mathbf{2.279\ mA}, \qquad I_E = (\beta + 1) I_B = \mathbf{2.298\ mA}$$

$$V_{CE} = V_{CC} - I_C R_C - I_E R_E = 18 - 7.522 - 2.298 = \mathbf{8.18\ V}$$

Active check: $V_{CE} = 8.18\ \text{V} > V_{CE,sat} = 0.2\ \text{V}$, so the BJT is in the active region. (Also $V_B = 3.00\ \text{V}$ and $V_C = 10.48\ \text{V}$, so the CB junction is reverse biased.)

(The approximate method gives $I_E = 2.43$ mA, about 6% high.)

(b)

$$r_e = \frac{26\ \text{mV}}{I_E} = \frac{26}{2.298} = \mathbf{11.31\ \Omega}, \qquad \beta r_e = 1.357\ \text{k}\Omega$$

$$Z_i = R_1 \parallel R_2 \parallel \beta r_e = 39\ \text{k} \parallel 8.2\ \text{k} \parallel 1.357\ \text{k} = \mathbf{1.131\ k\Omega}$$

$$Z_o = R_C \parallel r_o = 3.3\ \text{k} \parallel 50\ \text{k} = \mathbf{3.096\ k\Omega}$$

$$A_v = -\frac{R_C \parallel r_o}{r_e} = -\frac{3096}{11.31} = \mathbf{-273.7} \quad (-291.7 \text{ if } r_o \text{ is ignored})$$

(c)

$$A_{v_L} = -\frac{R_C \parallel r_o \parallel R_L}{r_e} = -\frac{1866}{11.31} = \mathbf{-165.0}$$

Check with the Thevenin output form: $A_v \dfrac{R_L}{R_L + Z_o} = -273.7 \times \dfrac{4.7}{7.796} = -165.0$.

$$A_{v_s} = \frac{Z_i}{Z_i + R_s} A_{v_L} = \frac{1131}{1731}(-165.0) = \mathbf{-107.8}$$

$$v_o = 107.8 \times 10\ \text{mV} = \mathbf{1.08\ V\ peak}\ (180^\circ \text{ out of phase with } v_s)$$

Which costs more gain? R_L does. It cuts the gain by the factor $165/274 = 0.60$, because R_L (4.7 kΩ) is not much bigger than Z_o (3.1 kΩ). R_s cuts it by

$1131/1731 = 0.65$, because Z_i (1.13 kΩ) is only about twice R_s. The two losses are similar, and R_L is slightly worse.

Marks (13):

- a. 1: the $\beta R_E \geq 10R_2$ check, stated. 1: V_{Th} and R_{Th}. 1: the I_B formula with $(\beta + 1)R_E$. 1: I_C and I_E. 1: V_{CE} with the active-region statement.
- b. 1: r_e from I_E (not I_C). 1: Z_i. 1: Z_o including r_o. 1: A_v with its negative sign.
- c. 1: A_{v_L}. 1: A_{v_s}. 1: v_o. 1: the R_s versus R_L comparison with reasons.

COMMON MISTAKES:

- Using R_{Th} with β instead of $(\beta + 1)$ in the I_B denominator. This is a small error, but it is a method error.
- Including R_E in Z_i or A_v when it is bypassed.
- Using $Z_i = \beta r_e$ alone and forgetting $R_1 \parallel R_2$.
- Getting A_{v_s} by dividing by R_s alone instead of by the divider $Z_i/(Z_i + R_s)$.
- Leaving out the minus sign. CE inverts.

REREAD: E5 (exact versus approximate VDB), E6 (CE bypassed, r_e model), E7 (loading with R_L and R_s).

Question 4: Emitter follower [9 marks]

(a) Base loop: $V_{CC} = I_B R_B + V_{BE} + (\beta+1) I_B R_E$.

$$I_B = \frac{12 - 0.7}{220\ \text{k} + 101(2.2\ \text{k})} = \frac{11.3}{442.2\ \text{k}} = \mathbf{25.55\ \mu A}$$

$I_E = (\beta+1) I_B = \mathbf{2.581\ mA}$, $V_E = I_E R_E = \mathbf{5.68\ V}$, $V_{CE} = 12 - 5.68 = \mathbf{6.32\ V}$

(b) $r_e = 26/2.581 = \mathbf{10.07\ \Omega}$.

No load: $Z_b = \beta r_e + (\beta+1) R_E = 1.007\ \text{k} + 222.2\ \text{k} = 223.2\ \text{k}\Omega$, so

$$Z_i = R_B \parallel Z_b = 220\ \text{k} \parallel 223.2\ \text{k} = \mathbf{110.8\ k\Omega}$$

With R_L: the emitter sees $R_E \parallel R_L = 2.2\ \text{k} \parallel 1\ \text{k} = 687.5\ \Omega$.

$$Z_b = 1.007\ \text{k} + 101(687.5) = 70.44\ \text{k}\Omega, \qquad Z_i = 220\ \text{k} \parallel 70.44\ \text{k} = \mathbf{53.4\ k\Omega}$$

(The course approximation $Z_b \cong \beta(r_e + R_E \parallel R_L)$ gives 52.96 kΩ. Accept it.)

(c) Looking back into the emitter, the base-side resistance is $R_s' = R_s \parallel R_B = 1\ \text{k} \parallel 220\ \text{k} = 995.5\ \Omega$. It is reflected down by β:

$$Z_o = R_E \parallel \left(r_e + \frac{R_s'}{\beta} \right) = 2.2\ \text{k} \parallel (10.07 + 9.955) = \mathbf{19.8\ \Omega}$$

(If you ignore R_s you get $R_E \parallel r_e = 10.0\ \Omega$. Give 1 of the 2 marks.)

(d)

$$A_{v_L} = \frac{R_E \parallel R_L}{R_E \parallel R_L + r_e} = \frac{687.5}{697.6} = \mathbf{0.986}$$

$$A_{v_s} = \frac{Z_i}{Z_i + R_s} A_{v_L} = \frac{53.36}{54.36}(0.986) = \mathbf{0.967} \quad \text{(in phase)}$$

(e) It has a high input impedance (53 kΩ) and a low output impedance (about 20 Ω). It can therefore drive a low-resistance load from a high-resistance source with almost no loss of signal: it is a buffer, and its current gain is large.

Marks (9): (a) 1: the I_B loop with $(\beta+1)R_E$. 1: I_E, V_E and V_{CE}. (b) 1: r_e and the no-load Z_i. 1: the loaded Z_i using $R_E \parallel R_L$. (c) 1: the R_s'/β reflection. 1: 19.8 Ω. (d) 1: A_{v_L}. 1: A_{v_s}. (e) 1: buffer, with high Z_i and low Z_o both stated.

> **COMMON MISTAKES:**

- Forgetting that R_L changes Z_i for a follower (Trap 2 in E7).
- Quoting $Z_o = R_E \parallel r_e$ when the source has resistance (Trap 1 in E7).
- Getting the direction of the reflection wrong. Emitter resistance looks $(\beta + 1)$ times bigger from the base. Base resistance looks β times smaller from the emitter.

REREAD: E5 (emitter-bias DC loop), E7 (section 1.4, emitter follower).

Question 5: JFET self-bias [8 marks]

(a) Bias line: $V_{GS} = -I_D R_S$. Substitute it into Shockley's equation, working in mA and kΩ:

$$I_D = 10\left(1 - \frac{-0.68 I_D}{-4}\right)^2 = 10(1 - 0.17 I_D)^2$$

Expanding gives $10 - 3.4 I_D + 0.289 I_D^2 = I_D$, so

$$0.289 I_D^2 - 4.4 I_D + 10 = 0 \quad \Rightarrow \quad I_D = \frac{4.4 \pm \sqrt{4.4^2 - 4(0.289)(10)}}{2(0.289)} = 12.44 \text{ mA or } 2.781 \text{ mA}$$

- Root 1: $I_D = 12.44$ mA gives $V_{GS} = -8.46$ V. This is beyond $V_P = -4$ V, where the JFET is off, and it is also more than I_{DSS}. **Reject it.**
- Root 2: $I_D = 2.781$ mA gives $V_{GS} = -1.89$ V, which lies between V_P and 0. **Accept it.**

$$\mathbf{I_{DQ} = 2.78\ mA}, \qquad \mathbf{V_{GSQ} = -1.89\ V}$$

(b)

$$V_S = I_D R_S = \mathbf{1.89\ V}, \qquad V_D = 20 - 2.781(2.2) = \mathbf{13.88\ V}, \qquad V_{DS} = \mathbf{11.99\ V}$$

Saturation check: $V_{DS} = 12.0\ \text{V} \geq V_{GS} - V_P = -1.89 + 4 = 2.11$ V. The JFET is pinched off (saturation), so Shockley's equation was valid.

(c)

$$g_{m0} = \frac{2 I_{DSS}}{|V_P|} = \frac{20}{4} = 5\ \text{mS}, \qquad g_m = 5\left(1 - \frac{-1.89}{-4}\right) = \mathbf{2.64\ mS}$$

$$r_d = \frac{1}{y_{os}} = \mathbf{50\ k\Omega}, \qquad Z_i = R_G = \mathbf{1\ M\Omega}, \qquad Z_o = r_d \parallel R_D = \mathbf{2.11\ k\Omega}$$

$$A_v = -g_m(r_d \parallel R_D) = -2.637\ \text{mS} \times 2.107\ \text{k}\Omega = \mathbf{-5.56}$$

(d) With R_S unbypassed, the signal is $v_i = v_{gs} + g_m v_{gs} R_S$ (because $i_d = g_m v_{gs}$ flows through R_S), and $v_o = -g_m v_{gs} R_D$. So

$$A_v = -\frac{g_m R_D}{1 + g_m R_S} = -\frac{2.637(2.2)}{1 + 2.637(0.68)} = \mathbf{-2.08}$$

(Keeping r_d, the full formula $-g_m R_D / [1 + g_m R_S + (R_D + R_S)/r_d]$ gives -2.03.)

Marks (8): (a) 1: the correct quadratic. 1: both roots. 1: the correct root, with the reason. (b) 1: V_S, V_D and V_{DS}. 1: the saturation check. (c) 1: g_m. 1: Z_i, Z_o and A_v. (d) 1: the derivation and -2.08.

COMMON MISTAKES:

- Getting a sign wrong inside the bracket. V_{GS}/V_P is a ratio of two negatives, so it is positive: $(1 - 0.17I_D)$, not $(1 + 0.17I_D)$.
- Taking the larger root because it "looks like a bigger current".
- Using g_{m0} instead of g_m at the Q-point.
- In (d), forgetting the $(1 + g_m R_S)$ factor. It is the FET version of $R_C/(r_e + R_E)$.

REREAD: E10 (self-bias, choosing the root), E11 (g_m, CS gain), E12 (unbypassed R_S).

Question 6: E-MOSFET voltage-divider bias [7 marks]

(a)

$$k = \frac{I_{D(on)}}{\left(V_{GS(on)} - V_{GS(Th)}\right)^2} = \frac{5\ \text{mA}}{(8-3)^2} = \mathbf{0.2\ mA/V^2}$$

(b)

$$V_G = \frac{6.8}{16.8} \times 24 = \mathbf{9.714\ V}, \qquad V_{GS} = V_G - I_D R_S = 9.714 - I_D\ (\text{mA, k}\Omega)$$

Substitute into $I_D = k(V_{GS} - 3)^2$:

$$I_D = 0.2(6.714 - I_D)^2 \quad \Rightarrow \quad 0.2I_D^2 - 3.686 I_D + 9.016 = 0$$

$I_D = 15.52\ \text{mA}$ (gives $V_{GS} = -5.81\ \text{V}$) or $I_D = 2.904\ \text{mA}$ (gives $V_{GS} = 6.81\ \text{V}$)

The first root has $V_{GS} < V_{GS(Th)}$, so the device would be off and no current could flow. **Reject it.**

$$\mathbf{I_{DQ} = 2.90\ mA}, \qquad \mathbf{V_{GSQ} = 6.81\ V}$$

(c)

$$V_{DS} = V_{DD} - I_D(R_D + R_S) = 24 - 2.904(3) = \mathbf{15.29\ V}$$

$$V_{DS(sat)} = V_{GS} - V_{GS(Th)} = 3.81\ \text{V} < 15.29\ \text{V}$$

The MOSFET is in saturation, so the square law was valid.

(d)

$$g_m = 2k(V_{GSQ} - V_{GS(Th)}) = 2(0.2)(3.81) = \mathbf{1.52\ mS}, \qquad A_v \approx -g_m R_D = \mathbf{-3.05}$$

Marks (7): (a) 1. (b) 1: V_G and the bias line. 1: the quadratic and both roots. 1: the right root, with the reason. (c) 1: V_{DS}. 1: the saturation comparison. (d) 1: g_m and A_v.

COMMON MISTAKES:

- Using Shockley's equation (with I_{DSS} and V_P) for an E-MOSFET.
- Taking $k = I_{D(on)}/V_{GS(on)}^2$, which forgets to subtract the threshold.
- Picking the root without checking it against $V_{GS(Th)}$.
- Leaving out the saturation check. "Confirm" parts carry a mark for the stated comparison.

REREAD: E9 (E-MOSFET characteristic), E10 (E-MOSFET voltage-divider bias), E11 ($g_m = 2k(V_{GS} - V_T)$).

Question 7: Two-stage cascade, JFET into BJT [8 marks]

(a)

$$g_{m0} = \frac{2(8)}{4} = 4\ \text{mS}, \qquad g_{m1} = 4\left(1 - \frac{-1.5}{-4}\right) = \mathbf{2.5\ mS} \qquad (I_{DQ} = 3.125\ \text{mA})$$

$$r_{e2} = \frac{26\ \text{mV}}{2.5\ \text{mA}} = \mathbf{10.4\ \Omega}, \qquad \beta r_{e2} = 1.56\ \text{k}\Omega$$

$$Z_{i2} = R_1 \parallel R_2 \parallel \beta r_{e2} = 33\ \text{k} \parallel 10\ \text{k} \parallel 1.56\ \text{k} = \mathbf{1.296\ k\Omega}$$

(b) Each stage is loaded by whatever follows it.

$$A_{v1} = -g_{m1}(R_{D1} \parallel Z_{i2}) = -2.5\ \text{mS} \times (3.3\ \text{k} \parallel 1.296\ \text{k}) = -2.5 \times 0.9306 = \mathbf{-2.33}$$

$$A_{v2} = -\frac{R_{C2} \parallel R_L}{r_{e2}} = -\frac{2.2\ \text{k} \parallel 4.7\ \text{k}}{10.4} = -\frac{1498.6}{10.4} = \mathbf{-144.1}$$

$$A_{vT} = A_{v1}A_{v2} = (-2.327)(-144.1) = \mathbf{+335}$$

(c) $Z_i = R_G = \mathbf{1\ M\Omega}$ and $Z_o = R_{C2} = \mathbf{2.2\ k\Omega}$.

$$A_{v_s} = \frac{R_G}{R_G + R_{sig}} A_{vT} = \frac{1000}{1010}(335.3) = \mathbf{332}$$

$$v_o = 332 \times 2\ \text{mV} = \mathbf{0.664\ V\ peak},\ \text{in phase with } v_s\ (\text{two inversions})$$

(DC sanity check, not required: stage 1 has $V_{DS} = 15 - 3.125(3.3 + 0.48) = 3.19$ V, which is at least $V_{GS} - V_P = 2.5$ V, so it is pinched off.)

(d) The BJT stage's input impedance (1.3 kΩ) is much lower than R_{D1} (3.3 kΩ), so it shunts the JFET's drain load and the gain collapses from -8.25 to -2.33. Fix: put a buffer (an emitter follower or source follower) between the stages, or redesign stage 2 for a higher Z_i, for example with a partly unbypassed R_E.

Marks (8): (a) 1: g_{m1}. 1: r_{e2} and Z_{i2}. (b) 1: A_{v1} loaded by Z_{i2}. 1: A_{v2} loaded by R_L. 1: A_{vT} positive. (c) 1: Z_i and Z_o. 1: A_{v_s}, v_o and the phase. (d) 1.

COMMON MISTAKES:

- Multiplying the **no-load** gains: $(-8.25)(-211.5) = 1745$, which is five times too big. This is the most common cascade error.
- Loading stage 1 with R_L, or loading stage 2 with Z_{i1}.
- Forgetting that with $R_G = 1$ MΩ the source resistance hardly matters. The factor is 0.99.
- Giving a negative overall gain. Two inverting stages make it non-inverting.

REREAD: E8 (cascades and loaded-gain method), E11 (CS gain), E12 (multistage FET).

Question 8: CMRR and the differential amplifier [7 marks]

(a) (WS10 Q2 method.)

$$V_d = V_{i1} - V_{i2} = 0.1\ \text{mV}, \qquad V_c = \tfrac{1}{2}(V_{i1} + V_{i2}) = 1.0\ \text{mV}$$

$$V_o = A_d V_d \left(1 + \frac{1}{\text{CMRR}}\frac{V_c}{V_d}\right) = 4000(0.1\ \text{mV})\left(1 + \frac{10}{\text{CMRR}}\right) = 0.4\ \text{V}\left(1 + \frac{10}{\text{CMRR}}\right)$$

- i. 80 dB means CMRR $= 10^{80/20} = 10^4$: $V_o = 0.4(1.001) = \mathbf{400.4\ mV}$.
- ii. 40 dB means CMRR $= 100$: $V_o = 0.4(1.1) = \mathbf{440\ mV}$.

With a poor CMRR, the common-mode error is 10% of the output.

(b) Base at 0 V, so the emitters are at -0.7 V:

$$I_{tail} = \frac{-0.7 - (-12)}{15\ \text{k}} = 0.7533\ \text{mA}, \qquad I_E = \frac{I_{tail}}{2} = \mathbf{0.377\ mA}\ \text{each}$$

$$r_e = \frac{26}{0.3767} = \mathbf{69.0\ \Omega}, \qquad V_C = 12 - \alpha I_E R_C = 12 - 0.3729(10) = \mathbf{8.27\ V}$$

$$A_d = \frac{R_C}{2r_e} = \frac{10\ \text{k}}{138.1} = \mathbf{72.4}$$

$$A_c = \frac{\beta R_C}{r_i + 2(\beta+1)R_E} = \frac{100(10\ \text{k})}{6.90\ \text{k} + 202(15\ \text{k})} = \mathbf{0.329} \quad (\approx R_C/2R_E = 0.333)$$

$$\text{CMRR} = \frac{72.4}{0.329} = \mathbf{220} = 20\log_{10} 220 = \mathbf{46.8\ dB}$$

Marks (7): (a) 1: V_d and V_c. 1: the formula, or the $A_c = A_d/\text{CMRR}$ route. 1: both outputs. (b) 1: I_E, r_e and V_C. 1: A_d. 1: A_c. 1: CMRR as a ratio and in dB.

COMMON MISTAKES:

- Using the dB figure directly as the ratio (CMRR = 80). Convert it first: CMRR $= 10^{\mathrm{dB}/20}$.
- Using $V_c = V_{i1} + V_{i2}$ without the factor of one half.
- Giving each transistor the full tail current. It splits in two.
- Using $10\log_{10}$ for a voltage-gain ratio.

REREAD: E13 (differential amplifiers and CMRR; WS10 Q1–Q2).

Question 9: Op-amps [5 marks]

(a)

$$V_{o1} = \left(1 + \frac{R_f}{R_1}\right) V_1 = \left(1 + \frac{47}{4.7}\right)(20\ \mathrm{mV}) = 11 \times 20 = \mathbf{220\ mV\ rms}\ (\text{in phase})$$

(b)

$$V_o = -\left(\frac{100}{33} V_{o1} + \frac{100}{10} V_2\right) = -(3.030 \times 220 + 10 \times 50)\ \mathrm{mV} = -(666.7 + 500) = \mathbf{-1.167\ V\ rms}$$

That is 1.17 V rms, 180° out of phase with the inputs. Its peak is 1.65 V, well inside the ±15 V rails.

(c) $10\ \mathrm{k\Omega}$. Op-amp 2's inverting input is a virtual ground, so V_2 sees only its own series resistor.

Marks (5): (a) 1: the formula. 1: 220 mV. (b) 1: the summing formula with each signal's own gain. 1: -1.17 V. (c) 1.

COMMON MISTAKES:

- Using the inverting gain $R_f/R_1 = 10$ for op-amp 1. It is non-inverting, so the gain is 11.
- Dropping the minus sign, or adding the gains before multiplying.

REREAD: E13 (op-amp configurations; WS10 Q3–Q5).

Question 10: Power amplifiers [7 marks]

(a)

$$I_{BQ} = \frac{18 - 0.7}{1.2\ \text{k}} = 14.42\ \text{mA}, \qquad I_{CQ} = \beta I_{BQ} = \mathbf{576.7\ mA}$$

$$V_{CEQ} = 18 - 0.5767(20) = \mathbf{6.47\ V}$$

AC: $I_{c(p)} = \beta I_{b(p)} = 40(8\ \text{mA}) = 320$ mA peak, so $V_{c(p)} = 6.4$ V. This is just inside $V_{CEQ} = 6.47$ V, but it takes V_{CE} down to 0.07 V, slightly below $V_{CE,sat} = 0.2$ V, so the negative collector-voltage peaks just touch saturation (the undistorted limit is $(V_{CC} - 0.2)/R_C - I_{CQ} = 313$ mA). The clipping is about 2%, so use the ideal sinusoid formula and say it is marginal.

$$P_{o(ac)} = \frac{I_{c(p)}^2 R_C}{2} = \frac{0.32^2(20)}{2} = \mathbf{1.02\ W}$$

$$P_{i(dc)} = V_{CC} I_{CQ} = 18(0.5767) = \mathbf{10.38\ W}, \qquad \eta = \frac{1.024}{10.38} = \mathbf{9.9\%}$$

(This is well below the 25% ceiling for a series-fed class A stage. The Q-point is not centred, so the swing is limited by V_{CEQ}.)

(b) (WS10 Q10 method.)

$$P_{i(max)} = \frac{2V_{CC}^2}{\pi R_L} = \frac{2(900)}{16\pi} = \mathbf{35.8\ W}$$

$$P_{o(max)} = \frac{V_{CC}^2}{2R_L} = \frac{900}{32} = \mathbf{28.1\ W}$$

$$\eta_{max} = \frac{28.125}{35.81} = \frac{\pi}{4} = \mathbf{78.5\%}$$

(Not asked: the two transistors together dissipate at most $2V_{CC}^2/(\pi^2 R_L) = 11.4$ W. That maximum occurs at $V_{L(p)} = 0.636 V_{CC}$, not at full output.)

Marks (7): (a) 1: I_{CQ} and V_{CEQ}. 1: P_o, using peak$^2/2$ or rms^2 consistently. 1: P_i. 1: η. (b) 1 each for (i), (ii) and (iii).

COMMON MISTAKES:

- Using the peak current as if it were rms in $P_o = I^2 R$. That doubles the power.
- Taking P_i as P_o plus something guessed. Class A input power is simply $V_{CC} I_{CQ}$.

- In class B, using V_{CC}^2/R_L for P_o. The factor of 2 for a sine wave is missing.

REREAD: E13 (power amplifiers; WS10 Q6–Q10).

Score interpretation

Write your marks in this table first:

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Total
/10	/8	/13	/9	/8	/7	/8	/7	/5	/7	/82

Also add up your score in two blocks: **diodes and BJT** (Q1 I–V, Q2, Q3, Q4: 35 marks) and **FET, multistage, op-amps and power** (Q1 VI–X, Q5–Q10: 47 marks). The weaker block is where your last two days go.

Score	What it means	What to do in the two days left
66–82 (80%+)	Exam-ready. The methods are automatic.	Do EX2 (mock B) timed tomorrow. Spend the rest of the time on any question where you lost 2 or more marks. Read the traps lists in E8 and E12 once more. Finish your reference sheet (REFE).
49–65 (60–79%)	Pass with a margin. Some method gaps or slips.	For each question below 60%, reread its listed pack's method section (section 2) and redo the question with the guide closed. Then sit EX2. Check that your reference sheet has every formula you looked up today.
33–48 (40–59%)	Borderline. You have the ideas but not the full playbooks.	Focus on the canonical, high-mark types: Q3 (E5–E7), Q5 (E10–E11) and Q7 (E8). Rework the worked examples in those packs. Then do only Q3, Q5 and Q7 of EX2, timed. Make sure the formula table above is on your printed reference.
Below 33 (under 40%)	Not yet safe.	Do not try to cover everything. Learn four playbooks well: VDB DC to AC (E5, E7), JFET self-bias with the root choice (E10), the cascade loaded-gain rule (E8), and class B power (E13). Each one is worth 7–13 marks. Redo this paper's Q3, Q5, Q7 and Q10 the next day without notes on the method.

Patterns worth checking whatever your total:

- **Lost marks on Q1, VI–X:** the conceptual FET, op-amp and power material is under-revised. Reread E9 and E13 section 1.
- **Right method, wrong numbers on Q3–Q7:** usually units. Work in mA and kΩ throughout. Then $V = mA \times k\Omega$, g_m is in mS, and $g_m \times k\Omega$ is dimensionless.
- **Ran out of time:** you spent too long on Q3. It is the canonical question, so practise it until it takes 15 minutes.
- **Picked the wrong root in Q5 or Q6:** write the check line every time: "reject: $|V_{GS}| > |V_P|$" or "reject: $V_{GS} < V_{GS(Th)}$".